# Hansol Suh

Last updated: September 14, 2026

hansolsuh.com hansol.suh.123@gmail.com
GitHub: hansolsuh GitLab: hsuh

## Education

**Georgia Institute of Technology** Atlanta, GA

**MSc, Computational Science and Engineering** Fall 2022

PhD studies, Computational Science and Engineering: Spring 2018–Fall 2022. Program concluded with an MSc. Advisor: Tobin Isaac. Research: numerical optimization and HPC.

**BSc, Civil Engineering, Highest Honor** August 2011–July 2016

## Technical Skills

**Programming:** C, C++, Python, Julia, Java, MATLAB; CUDA and HIP.

**Methods and tools:** MPI, OpenMP, Unix/Linux, Git, machine learning, parallel computing.

## Publications

### Peer-reviewed publications

**Verification of PETSc with CIVL Using LLM-Generated ACSL Contracts and Deterministic Driver Generation**

**H. Suh**, J. Hückelheim, and S. Siegel.

Correctness 2026: 10th International Workshop on Software Correctness for HPC Applications (SC26). Accepted; workshop scheduled for November 16, 2026.

**PETSc/TAO Developments for GPU-Based Early Exascale Systems**

R. T. Mills, M. F. Adams, S. Balay, J. Brown, J. Faibussowitsch, T. Isaac, M. G. Knepley, T. Munson, **H. Suh**, S. Zampini, H. Zhang, and J. Zhang.

The International Journal of High Performance Computing Applications. 2025.

doi:10.1177/10943420241303710

**Evaluation of a Minimally Synchronous Algorithm for 2:1 Octree Balance**

**H. Suh** and T. Isaac.

SC20: International Conference for High Performance Computing, Networking, Storage and Analysis. November 2020.

### Posters

**Discrete Adjoints in PFLOTRAN for Physics-Constrained Machine Learning**

**H. Suh** et al.

AGU Annual Meeting 2026, San Francisco, CA. Submitted.

### Preprints and manuscripts

**A Portable and Versatile Limited-Memory BFGS Implementation in PETSc/TAO**

**H. Suh**, T. Isaac, A. Dener, T. Munson, H. Zhang, and R. T. Mills.

arXiv:2608.00196. 2026.

**Variable Metric Davis–Yin Splitting**

**H. Suh** and T. Isaac.

Manuscript.

**A Heuristic for Parameter Selection in Adaptive Relaxed ADMM Based on Numerical Ranges**

T. Isaac and **H. Suh**.

Manuscript.

## Technical Reports and Manuals

**PETSc/TAO Users Manual, Revision 3.25**

S. Balay et al. (including **H. Suh**).

ANL-21/39 Rev. 3.25. Argonne National Laboratory. 2026.

**TAO Users Manual, Revision 3.15**

A. Dener, A. Denchfield, **H. Suh**, T. Munson, J. Sarich, S. Wild, S. Benson, and L. C. McInnes.

ANL/MCS-TM-322 Rev. 3.15. Argonne National Laboratory. March 2021.

## Experience

**Argonne National Laboratory** Lemont, IL

**Predoctoral Appointee** March 2023–Present

- Develop dense BFGS implementations for GPUs in PETSc/TAO.
- Develop proximal optimization solvers for PETSc/TAO.
- Supervisors: Tobin Isaac (former); Richard Mills (current).

**Summer Research Appointments** 2019, 2021, 2022

- Research Aide, Technical (June–August 2022): developed Anderson-accelerated three-operator splitting and hierarchical, composable PETSc/TAO solvers. Supervisor: Todd Munson.
- Research Aide (May–August 2021): implemented variable-metric three-operator splitting in PETSc/TAO and evaluated the implementation in an HPC setting. Supervisor: Alp Dener.
- Givens Associate (May–August 2019): implemented ADMM and a variant in PETSc/TAO and evaluated their performance in an HPC setting. Supervisor: Alp Dener.

**Georgia Institute of Technology** Atlanta, GA

**Graduate Research Assistant** May 2018–December 2022

Research with Tobin Isaac in PETSc development, numerical optimization and analysis, p4est, and HPC.

**Graduate Student** August 2016–May 2018

**Republic of Korea Army** Icheon, South Korea

Anti-Aircraft Gunner; Armory Department. November 2013–August 2015

## Grants & Funding

**CS2: Modular Verification of Scientific Software** May 2025–May 2029

**Using Model Checking, Symbolic Execution, and Abstraction**

Senior Personnel, NSF/DOE Correctness for Scientific Computing Systems (CS2), NSF 24-571. Award: $3,200,000. PIs: Stephen Siegel (University of Delaware), Jan Hückelheim (Argonne National Laboratory).

**AI4Soil** Summer 2026

A Deployable Artificial Intelligence Soil Prototype for Improving Land–Atmosphere Prediction. ANL LDRD Seed project; collaborator with Joshua Minai. Award: $25,000.

## Projects

**Terrestrial Wetland Function and Resilience Science Focus Area** Summer 2025–Present

Developing adjoint-based sensitivity-analysis capabilities in PFLOTRAN for subsurface reactive transport, and physics-constrained machine-learning methods for wetland systems.

**TaoTerm**

PIONEER thrust: Computationally Efficient Composable Solvers for PETSc/TAO. Developed composable objects for the TAO optimizer, enabling flexible, on-the-fly modification of optimization objectives.

## Presentations

**TaoTerm: Composable Objective Function Terms**

**H. Suh**.

Consortium for the Advancement of Scientific Software BoF Days, online. February 11, 2026.

Slides

**Dense Broyden–Fletcher–Goldfarb–Shanno (BFGS)**

**H. Suh**.

PETSc User Meeting 2025, Buffalo, NY. May 20, 2025.

Slides

**An Introduction to PETSc/TAO**

**H. Suh**.

ALCF Hands-On HPC Workshop, Lemont, IL. October 31, 2024.

Slides

**Mini Tutorial: TaoADMM**

**H. Suh**.

PETSc User Meeting 2023, Chicago, IL. June 7, 2023.

Slides

**An Asynchronous Algorithm for 2:1 Octree Balance**

**H. Suh** and T. Isaac.

SIAM Conference on Parallel Processing 2020, Seattle, WA. February 14, 2020.

**Using the PETSc/TAO ADMM Methods on GPUs**

**H. Suh**, A. Dener, T. Isaac, and T. Munson.

SIAM Conference on Parallel Processing 2020, Seattle, WA. February 13, 2020.

## Teaching

**Computational Data Analysis (ISYE 6740)** Fall 2021, Fall 2022

Teaching Assistant.

**Introduction to High Performance Parallel Computing (CX 4220)** Spring 2021

Teaching Assistant.

**High Performance Parallel Computing (CSE 6230)** Fall 2019

Teaching Assistant.